

Andrés Carvajal

Atlanta, GA

andres_csuares@live.com — (470) 926-3159

[LinkedIn](#) — [Website](#) — [GitHub](#)

SUMMARY

AI Engineer with 3+ years of experience building production-grade AI systems using LLMs, RAG architectures, and diffusion models. Specialized in designing scalable pipelines, multi-agent systems, and cloud-based AI infrastructure on AWS and GCP. Strong background in Python, backend development, and end-to-end deployment of AI-powered products. Available immediately and open to contract or full-time roles.

EXPERIENCE

RX-M

AI Engineer / Consultant

Sep. 2025 – Mar. 2026

Remote, USA

- Designed and deployed **LLM-based systems** using **LangChain** and **LangGraph**, enabling multi-agent orchestration and tool-based reasoning workflows.
- Built production **RAG pipelines** integrating **Pinecone** and **Qdrant**, improving retrieval accuracy and response relevance in enterprise applications.
- Developed **AI automation pipelines** for document processing and legal workflows, extracting structured data from PDFs, contracts, and IDs.
- Integrated APIs from **OpenAI**, **Claude**, and **Gemini** into scalable backend systems deployed on **AWS** and **GCP**.
- Led training programs on **Generative AI systems**, covering agents, evaluation strategies, and production deployment.

RandomAI

AI Engineer

Jan. 2025 – Sep. 2025

Remote, USA

- Architected a **serverless AI platform** combining frontend, backend, and AI pipelines using **Firebase** and cloud infrastructure.
- Built end-to-end **image generation pipelines** using **Stable Diffusion (SDXL)** and **FLUX**, deployed via **Replicate** and **Runpod**.
- Automated product creation workflows integrating **Shopify** and **Printful**, reducing manual operations and improving scalability.
- Developed AI-powered **content generation pipelines** for marketing, enabling automated image generation and campaign deployment.
- Integrated **cloud storage and processing pipelines on GCP**, including image preprocessing, background removal, and upscaling.

DeepMake

Machine Learning Engineer

Apr. 2023 – Nov. 2024

Remote, USA

- Developed **real-time AI inference systems** for video segmentation and VFX using **PyTorch**, **Stable Diffusion**, and **GroundingDINO-SAM**.
- Built scalable **FastAPI services** with dynamic GPU allocation and autoscaling on **AWS EC2 Spot Instances**.
- Implemented **ML pipelines** for training and deploying models, including fine-tuning diffusion models and super-resolution systems.

- Designed **distributed inference architecture** with monitoring and logging using **Sentry**.
- Optimized model performance and reduced latency for production use cases in creative applications.

CAREL USA

Software Developer & Automation Engineer

Jul. 2022 – Apr. 2023

Roswell, GA

- Developed **automation systems** and control logic for HVAC systems using Structured Text PLCs.
- Built **data pipelines and dashboards** for energy analytics and forecasting.
- Integrated IoT telemetry systems and improved operational efficiency through automation.

EDUCATION

Universidad de Los Andes (UNIANDES)

M.Sc. in Civil Engineering (Geotechnical)

Bogotá, Colombia

Jun. 2021

Universidad Pontificia Bolivariana (UPB)

B.Sc. in Civil Engineering

Bucaramanga, Colombia

Dec. 2017

SKILLS

- **Programming:** Python, JavaScript (Node.js), Bash, SQL
- **AI / Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Diffusers, LLMs, Generative AI, Databricks ML, MLflow, Model Fine-tuning, Inference Optimization
- **LLM & Agentic Systems:** RAG (Retrieval-Augmented Generation), LangChain, LangGraph, Google ADK Stack, Agentic AI on Databricks/Azure, Prompt Engineering, Multi-Agent Systems, Tool Calling, Evaluation Pipelines
- **Generative AI:** Stable Diffusion (SDXL), FLUX, ControlNet, LoRA, Image Generation, Image-to-Image, Inpainting, Upscaling
- **Backend Engineering:** FastAPI, REST APIs, Microservices, API Design, Asynchronous Processing, Distributed Systems
- **Data Engineering:** Data Pipelines, ETL, Data Processing, Feature Engineering, Polars, Pandas, Databricks
- **Cloud & Infrastructure:** AWS (EC2, S3, Lambda), Azure, GCP (Vertex AI, Compute Engine, Cloud Storage), Databricks, Serverless Architectures
- **Vector Databases & Storage:** Pinecone, Qdrant, PostgreSQL, BigQuery
- **MLOps & Deployment:** Docker, CI/CD, MLflow, Model Deployment, Monitoring, GPU Workloads, Scaling Systems
- **Tools:** Git, Linux/Unix, n8n, Supabase, Replicate, Runpod